Systematic Part: $\beta_0 + \beta_1 Xi$ or explained part and Random Part: U_i unexplained part in a regression. U_i represents the role of omitted variables in specifying a regression relationship of Y on X. Hence, the U_i cannot and should not be ignored.

Assumptions of the Linear Regression Model

The Linear regression model is based on certain assumptions

1. Some of them refer to the distribution of the random variable .
2. Some of them refer to the relationship between U_i and the explanatory variables (x_1, x_2, x_3 given in the above example).
3. Some of them refer to the relationship between U_i the explanatory variables themselves.

Assumptions about the distribution of the values of are called stochastic assumptions of ordinary least squares (OLS). Assumptions relating to the relationship between U_i and explanatory variables and relating to the relationship among the explanatory variables are called other assumptions.

Assumptions

1. "U" is a random real variable. That is "U" may assume positive, negative or zero values. Hence the mean of the "U" will be zero.
2. The variance of "U" is constant for all values of "U"
3. The "U" has a normal distribution.
4. The Covariances of any U_i with any other U_j are equal to zero
5. "U" is independent of explanatory variable (s)
6. Explanatory variables are measured without error.
7. The explanatory variables are not perfectly linearly correlated.
8. The variables are correctly aggregated.
9. The relationship is correctly identified and specified.
10. Parameters are linear.

12.15 Official Statistics

Official Statistics are statistics published by government agencies or other public bodies such as international organizations. They provide quantitative or qualitative information on all major areas of citizens' lives. Official Statistics make information on economic and social development accessible to the public, allowing the impact of government policies to be assessed, thus improving accountability.

The Ministry of Statistics and Programme Implementation (MOSPI) came into existence as an Independent Ministry in 1999 after the merging of the Department of Statistics and the Department of Programme Implementation.

The Ministry has two wings, Statistics and Programme Implementation.

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